MILITARY SPECIFICATION

SWITCHBOARD, TELETYPEWRITER SIGNAL DISTRIBUTION,
SB-3959/BGC

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for a nonsecure teletype signal distribution switchboard, hereinafter referred to as the switchboard, that connects cryptographic devices to radio transmitters and receivers or audio tone converters.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-S-901	Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment And Systems, Requirements For
MIL-C-5015	Connectors, Electrical, Circular Threaded, AN Type, General Specification For
MIL-E-15090	Enamel, Equipment, Light-Gray (Formula No. 111)
MIL-E-16400	Electronic, Interior Communication And Navigation Equipment, Naval Ship And Shore: General Specification For
MIL-E-17555	Electronic And Electrical Equipment, Accessories, And Repair Parts, Packaging And Packing Of
MIL-C-26482	Connectors, Electrical (Circular, Miniature, Quick Disconnect Environment Resisting), Receptacles And Plugs, General Specification For
MIL-C-28777/1	Cable Assembly, Electronic Test Equipment, 3 Wires, 3 Prong, 125 Volts, Molded Grounding Plug Connector And MIL-C-5015 Connector With Cable Clamp, 60 Hertz

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Electronic Systems Command (ELEX-8111), Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

FSC 5811

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STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures And Tables For Inspection By Attributes
MIL-STD-108	Definitions Of And Basic Requirements For Enclosures For Electric And Electronic Equipment
MIL-STD-109	Quality Assurance Terms And Definitions
MIL-STD-188	Military Communication System Technical Standards
MIL-STD-189	Racks, Electrical Equipment, 19-Inch And Associated Panels
MIL-STD-45[illegible]	Standard General Requirements For Electronic Equipment
MIL-STD-461	Electromagnetic Emission And Susceptibility Requirements For The Control Of Electromagnetic Interference
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement Of
MIL-STD-471	Maintainability Verification/Demonstration/ Evaluation
MIL-STD-781	Reliability Design Qualification And Production Acceptance Tests: Exponential Distribution
MIL-STD-810	Environmental Test Methods
MIL-STD-965	Parts Control Program
MIL-STD-1378	Requirements For Employing Standard Electronic Modules
MIL-STD-1472	Human Engineering Design Criteria For Military Systems, Equipment And Facilities

(Copies of specifications, standards, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

NATIONAL FIRE PROTECTION ASSOCIATION

NFPA 70-1978	National Electrical Code

(Application for copies should be addressed to the National Fire Protection Association, 470 Atlantic Avenue, Boston, MA 02210.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The switchboard shall be in accordance with MIL-E-16400, except as hereinafter specified.

3.1.1 First article. When specified, a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.2 Description. The switchboard shall have a transmit section and a receive section. The transmit section shall have the capability of switching two inputs to five outputs. The receive section shall switch six inputs to four outputs.

3.2.1 Crosspoint diagram. The crosspoints that are required to be actuated shall be as shown in FIGURE 1.

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RECEIVE SECTION

SWITCH NUMBER	POSITION NUMBER						OUTPUT CONNECTOR NUMBER	OUTPUT EQUIPMENT
	1	2	3	4	5	6		
1	X	X		X	X	X	J1	KW-7(R)
2	X	X	X	X			J2	KWR-37
3	X	X	X	X	X	X	J3	SPARE
4	X	X	X	X	X	X	J4	SPARE
INPUT CONNECTOR NUMBER	J5	J6	J7	J8	J9	J10		
INPUT EQUIPMENT	URA-17A	URA-17B	WRR-7	CV-2460	WSC-3#1	WSC-3#2		

TRANSMIT SECTION

SWITCH NUMBER	POSITION NUMBER		OUTPUT CONNECTOR NUMBER	OUTPUT EQUIPMENT
	1	2		
1	X	X	J11	URT-23 (1)
2	X	X	J12	URT-23 (2)
3	X	X	J13	CV-2460
4	X	X	J14	WSC-3#1
5	X	X	J15	WSC-3#2
INPUT CONNECTOR NUMBER	J16	J17		
INPUT EQUIPMENT	KW-7 (T)	SPARE		

NOTE: SWITCH POSITION ZERO SHALL BE OFF.

FIGURE 1. Crosspoint diagram.

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3.2.2 Interface requirements. The switchboard shall be designed to operate in accordance with the telegraph and data lower level input and output (positive and negative 6 ±1 volt (V)) standard interface requirements of MIL-STD-188.

3.2.2.1 Input equipment impedance. The output impedance of an input equipment, used in low level circuits, shall not exceed 100 ohms for currents whose magnitude is less than 0.01 ampere.

3.2.2.2 Output equipment resistance. The minimum input resistance of an output equipment, used in low level circuits, shall be 6 kilohms (kohms). The maximum total input capacitance of the device shall be 2500 picofarads.

3.2.2.3 Current loops. The switchboard shall be capable of supplying either 60-milliampere (mA) or 20-mA level signaling.

3.3 Characteristics.

3.3.1 Equipment selection. Input/output (I/O) device connections shall be selected by thumbwheel switches located on the front panel. Each switch shall represent an output device, and the positions on a switch shall represent the input devices that may be connected to the output device. Interconnection shall be accomplished by 10-position rotary switches.

3.3.2 Status displays. When an I/O connection has been selected, the name of the selected input device shall be displayed on the screen of a rear projection device mounted above the thumbwheel switch corresponding to the particular output device. Only the names of the input devices that may be connected to an output device by a particular switch (as indicated by the valid cross-points shown in FIGURE 1) shall appear on the corresponding rear projection display. The signal activating the display shall originate with each switch.

3.3.3 Display nomenclature. The name of the input device selected by a thumbwheel switch shall appear on two lines on the rear projection device. The required nomenclature appearing on the displays when the thumbwheel switches are in the indicated positions shall be as shown in TABLE I.

TABLE I. Display nomenclature.

Receive section	
Position	Display
1	URA 17A
2	URA 17B
3	WRR 7
4	CV 2460
5	WSC-3 #1
6	WSC-3 #2
Transmit section	
Position	Display
1	KW 7 (T)
2	Spare

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3.3.4 Internal power supply requirements. Any power supply required for the operation of the unit shall be of modular design and be an integral part of the switchboard.

3.3.5 Monitor meter. A front panel monitor meter shall be provided which shall be able to show a) presence of a low-level signal (±6 V), or b) loop current level for each individual input or output specified.

3.3.6 Signal level selection. Signal level selection shall be made at any input or output by choosing a printed circuit (PC) card. Rewiring shall not be permitted.

3.3.6.1 PC card key. A moveable key shall be furnished so that only one type of card may be used in a slot without relocating the key.

3.3.6.2 Card identification. Each card shall be coded (for example, with colored card extractors) so that it can be ascertained which level of card is being used while the card is installed.

3.3.6.3 Card adjustment indicator. An indicator shall be located on the card which shall indicate the card being monitored by the meter (see 3.3.5).

3.3.7 Signal interruption. There shall be no interruption caused on an existing channel by switching another equipment to or through that source.

3.3.8 Power failure. When primary power is removed from the switchboard, the switch shall remain in the desired configuration. It is not expected that the signal will be transmitted with power off.

3.3.9 Physical characteristics.

3.3.9.1 Dimensions. The switchboard shall be housed in an enclosure with the maximum dimensions and weight shown in TABLE II.

TABLE II. Dimensions and weight.

Width		Depth		Height		Weight	
cm[1/]	(in.)[2/]	cm	(in.)	cm	(in.)	kilograms	(pounds)
48.26	(19)	50.8 (including rear connector)	(20)	26.67	(10.5)	21.4	(47.5)

1/ Centimeters.
2/ Inches.

When uncrated for installation, the switchboard shall be capable of passage through a doorway 50.8 cm (20 in.) wide by 96.52 cm (38 in.) high (reduced by rounded corners on a 25.4 cm (10 in.) radius) and through a circular tube 62.5 cm (25 in.) in diameter.

3.3.9.2 Mounting. The switchboard shall be capable of being mounted in a standard 48.26-cm (19-in.) electrical equipment rack as specified in MIL-STD-189. The switchboard shall also be provided with side-mounted pivot type slides and slide rails for ease of removing the equipment from the rack.

3.3.10 Indicators and controls. The switchboard shall be provided with the indicators and controls specified in 3.3.10.1 through 3.3.10.3.1. The design of controls and indicators, and their arrangement on operation and maintainer panels, shall conform to the criteria of MIL-STD-1472.

3.3.10.1 Primary power control. A switch shall be provided to connect operating power. The switch shall break all power conductors of the power circuit.

3.3.10.2 Primary indicator. When the primary power control is turned on, a green indicator, mounted near the switch, shall be illuminated. The indicator shall be connected to the load side of the switch and across the input power conductors.

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3.3.10.3 Primary power. The switchboard shall conform to the requirements of MIL-E-16400 for primary power supply line voltage and frequency. Primary power shall be 115 V ±10 percent, 60 hertz (Hz) ±5 percent, single-phase. The switchboard shall also conform to the requirements of MIL-E-16400 for transient voltage and frequency. The total power required to operate the switchboard shall not exceed 115 watts.

3.3.10.3.1 Power cable assembly. The input power cable assembly shall be Type I in accordance with MIL-C-28777/1. Power cable connection to the switchboard shall be from the rear.

3.3.11 Reliability.

3.3.11.1 Qualitative reliability requirements. The upper test mean-time-between-failure (MTBF) (θ_0 as defined by MIL-STD-781) shall be 2200 hours. The lower test MTBF (θ_1 as defined by MIL-STD-781) shall be 1100 hours.

3.3.11.2 Component reliability. All switches shall have a minimum life of 100,000 operations. All displays shall have a minimum life of 25,000 hours.

3.3.12 Maintainability.

3.3.12.1 Quantitative corrective maintenance. The switchboard shall have a mean-corrective-maintenance-time (M_{ct}), not exceeding 0.5 hour and a maximum-corrective-maintenance-time (M_{maxct}), 95th percentile, not exceeding 1.5 hours when corrective maintenance is accomplished at the organizational level of maintenance by the replacement of lowest subassemblies (modules, PC boards) and chassis-mounted parts (electronic, electrical, electromechanical, and mechanical parts). The corrective maintenance time includes localization, isolation, disassembly, interchange, reassembly, alignment and checkout for all corrective maintenance actions. The equipment is considered to be in a corrective maintenance mode whenever a failure or malfunction occurs, including a malfunction that allows degraded operation.

3.3.12.2 Subassembly (module) maintainability. Repairable modules shall have a M_{ct} of 2.0 hours and a M_{maxct} of 4.0 hours, 95th percentile, at the organizational, intermediate, or depot level maintenance, as appropriate. M_{ct} for module repair includes localization, isolation, disassembly, interchange, reassembly, alignment and checkout for all maintenance tasks. For purposes of maintainability, a module is considered the lowest replaceable subassembly (PC card) for which repair is accomplished by the replacement of piece parts.

3.3.12.3 Modular construction. Predeveloped modules from the Standard Electronic Modules (SEM) program module listing may be used. Contracting officer approval shall be obtained, before designing a SEM in accordance with MIL-STD-1378, for use in the switchboard.

3.3.13 Environmental conditions. The switchboard shall be designed to conform to the environmental requirements specified in 3.3.13.1.

3.3.13.1 Temperature. The switchboard shall operate under temperature ranges specified for Range 4 equipment in accordance with MIL-E-16400.

3.4 Design and construction. The design objectives shall be to achieve minimum size and maximum reliability while reducing time to repair through the best techniques of modular construction.

3.4.1 Parts control. The parts to be incorporated in the equipment shall be controlled in accordance with MIL-STD-965, Procedure I.

3.4.1.1 Enclosure. The enclosure shall be aluminum or steel and shall conform to the dripproof requirement (45 degrees) of the Dripproof paragraph of MIL-STD-108.

3.4.1.2 Front panel. The front panel shall be constructed to permit easy access to the rear of the components mounted on the panel and to components located within the enclosure without removal of the entire switchboard.

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3.4.1.3 Blown fuse indicators. Blown fuse indicating fuseholders shall be located on the front panel. A fuse or an overcurrent trip unit of a circuit breaker shall be connected in series with each ungrounded conductor as determined by the National Electrical Code (NFPA 70-1978) for grounded source power, and shall be located on the load side of the power switch.

3.4.1.4 Enclosure bonding. The enclosure shall be furnished with mounting holes for bonding-strap ground connectors. The mounting holes shall conform to the requirements of MIL-E-16400.

3.4.1.5 Exterior finish. The exterior finish shall conform to the requirements of MIL-E-15090, Class 2.

3.4.1.6 Connectors. The external connectors shall be in accordance with MIL-C-5015 or MIL-C-26482 and shall be located at the rear of the switchboard. Mating connectors shall be provided.

3.4.1.7 Cables. Cables for the I/O devices and for primary power shall not be provided; however, mating connectors for the I/O device connectors and the primary power connector shall be supplied.

3.4.1.8 Handles. Handles shall be provided on the front panel to ensure safe and easy removal of the switchboard from the equipment rack. Handles shall protrude farther than any switch, but not more than 5.00 cm (2 in.) from the front panel.

3.4.2 Electromagnetic interference (EMI) characteristics. The switchboard shall be self-compatible and shall conform to the requirements of MIL-STD-461 for Class A5 equipment, with the exception that RS03 shall be performed at 10 V per meter. If line-to-ground power line filters are used to conform to the requirements of MIL-STD-461, the total capacitance shall not exceed 0.1 microfarad (μF) for 60-Hz equipments, or 0.01 μF for 400-Hz equipments.

3.4.3 Legends

3.4.3.1 Output-device name tags. The output-device name tag that appears above each switch on the front panel shall be attached so as to permit easy replacement.

3.4.3.2 Input-equipment-position legend. The legend on the front panel that correlates input device names to switch positions shall be attached so as to permit easy replacement.

3.4.4 Human engineering. The switchboard shall conform to the human engineering operational and maintenance design criteria of MIL-STD-1472 and the requirements specified herein. Specifically, the rear projection displays and device name tags and labels shall be in accordance with MIL-STD-1472.

3.4.5 Leakage current. The leakage current of the switchboard shall not exceed 5 mA.

3.4.6 Color coding of conductors. All conductors, other than printed wiring, shall be color coded in accordance with MIL-E-16400.

3.4.7 Special maintenance items. Special tools (see 6.5) or items required for maintenance shall be provided.

3.4.8 Workmanship screen. The switchboard shall withstand a defect detection vibration screen of random type vibration at $0.04g^2$/Hz ±3 decibels (dB) from 80 Hz to 350 Hz (see 4.5.7.2.1). The switchboard shall withstand a temperature cycling screen to assure elimination of initial failures due to random defective components and marginal operation due to component log variations or poor assembly processes and procedures (see 4.5.7.2.2).

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shall be documented by each activity and be available to the user

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at any time. Verification will consist of, but is not limited to, a) surveillance of the operations to determine that practices, methods, and procedures of the written quality program are being properly applied, b) Government product inspection to measure quality of the product to be offered for acceptance, and c) Government inspection of delivered products to assure compliance with all inspection requirements of this specification. Failure of the contractor to promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been taken or until conformance of the product to prescribed criteria has been demonstrated.

4.1.2 Quality assurance terms and definitions. Quality assurance terms used in this specification shall be as defined in MIL-STD-109.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)

b. Quality conformance inspection (see 4.4)

1. Production inspection (Group A) (see 4.4.1)

2. Production control inspection (Group B) (see 4.4.2)

3. Environmental inspection (Group C) (see 4.4.3)

c. Reliability testing (see 4.5.3)

d. Maintainability demonstration (see 4.5.4)

e. Inspection of preparation for delivery (see 4.6)

4.3 First article inspection. Unless otherwise specified (see 6.2.1), one unit shall be required for first article inspection. First article inspection shall consist of all examination and testing necessary to determine compliance with the requirements of this specification. First article inspection shall include the tests specified in TABLE III.

4.4 Quality conformance inspection. Quality conformance inspection shall be as specified in 4.4.1 through 4.4.4 and as shown in TABLE III.

4.4.1 Production inspection (Group A). Production inspection shall be conducted on each switchboard offered for delivery. The inspection shall comprise such examinations and tests as will prove the workmanship and reveal omissions and errors of the production process, such as functional and performance tests at a limited number of points in the required range, tests which detect deviations from design, tests of control adjustments, and tests which detect hidden defects of materials. Production inspection shall include the examinations and tests specified in TABLE III (Group A).

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TABLE III. Examinations and tests.

Examination or test	Requirement paragraph	Test paragraph	First article inspection	Quality conformance inspection		
				Group A	Group B	Group C
Surface examination						
Weight	3.3.9.1	4.5	X	X		
Size	3.3.9.1	4.5	X	X		
Parts control	3.4.1	4.5	X	X		
Finish	3.4.1.5	4.5	X	X		
Marking	3.1, 3.4.10	4.5	X	X		
Safety	3.1	4.5	X	X		
Workmanship	3.1	4.5.7	X	X		
Performance						
Crosspoints switching	3.2.1	4.5.1	X			
Equipment impedance and resistance	3.2.2.1, 3.2.2.2	4.5.1	X			
Display readability	3.3.2, 3.3.3	4.5.1	X	X		
Supply line variation	3.3.10	4.5.1	X	X		
Temperature						
Low temperature	3.3.13.1	4.5	X		X	
High temperature	3.3.13.1	4.5	X		X	
Humidity	3.1	4.5	X			X
Workmanship screen	3.4.8	4.5.7.2	X	X		
Interchangeability	3.1	4.5.2	X			X
Thermal survey	3.1	4.5	X			
Shock	3.1	4.5	X			X
Vibration	3.1	4.5	X			X
Inclination	3.1	4.5	X			X
Enclosure	3.4.1.1	4.5	X		X	X
Steady-state voltage and frequency	3.3.10.3	4.5	X		X	
Transient voltage	3.3.10.3	4.5	X		X	
Transient frequency	3.3.10.3	4.5	X		X	
Spike voltage	3.3.10.3	4.5	X		X	
Power interruption	3.1, 3.3.7	4.5	X		X	
Leakage current	3.4.5	4.5.6	X		X	
Electromagnetic interference	3.4.2	4.5.5	X			X
Temperature cycling	3.4.8.2	4.5.7.2	X	X		
Reliability test and demonstration	3.3.11	4.5.3	X			
Maintainability demonstration	3.3.12	4.5.4	X			

4.4.2 Production control inspection (Group B). Production control inspection shall be conducted on a sampling basis as specified in 4.4.2.1. The inspection shall consist of the examinations and tests which encompass functional and performance tests throughout the entire range of operation; tests which will detect any deterioration of the design by wear of such items as dies, molds, and jigs, and by substitution of different parts; tests which detect deviations in the processing of the materials; tests to determine temperature rise produced in operation and ability of the switchboards to withstand the heat; tests of efficiency; and tests of the performance with other equipment in a system. The tests shall be performed on the complete switchboard as offered for delivery. Production control inspection shall include the examinations and tests specified in TABLE III (Group B).

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4.4.2.1 Sampling plan for Group B inspection. Switchboards submitted for Group B inspection shall be selected by the Government quality assurance representative and shall be representative of current production. Group B inspection, including sampling, shall conform to the procedures for small-sample inspection of MIL-STD-105, using the special inspection levels. Group B inspection shall be performed on inspection lots that have passed Group A inspection. The acceptable quality level shall be 6.5 percent defective based on all tests performed on the lot and the inspection level shall be S-3 for normal, tightened, and reduced inspection.

4.4.2.2 Rejected lots. If an inspection lot is rejected, the contractor may withdraw the lot from further inspection. The contractor may also rework a rejected lot to correct the defective switchboards and reinspect the lot using tightened inspection. Rejected lots shall be kept separate from new lots and shall not lose their identity.

4.4.3 Environmental inspection (Group C). Environmental inspection shall be conducted on a sampling basis as specified in 4.4.3.1. The inspection shall encompass environmental tests to prove the durability of the materials, parts, units, and the equipment as a whole; simulated service tests; tests of the effects of changes of environment (such as extremes of temperature and humidity, effect of salt air); and tests of the effects of shock, vibration, and inclination. Environmental inspection shall include the examinations and tests specified in TABLE III (Group C).

4.4.3.1 Sampling for environmental inspection. One switchboard shall be selected from each successive 50 switchboards, or fraction thereof, produced. The first sample switchboard shall be selected from the first month's production.

4.4.3.2 Nonconforming environmental sample switchboards. If a sample switchboard fails the inspection specified in 4.4.3, the contractor shall immediately investigate the cause of failure and shall implement the corrective action necessary to correct units of product which were manufactured under the same conditions, with the same materials, processes, and so forth. The switchboards shall then be retested.

4.4.4 Reinspection of production control and environmental test switchboards. Unless otherwise specified in the contract (see 6.2.1), sample switchboards which have been subjected to and have passed production control and environmental inspections may be accepted on the contract provided they are resubjected to, and pass, the production inspection specified in 4.4.1 after repair of all visible damage.

4.5 Test methods. Unless otherwise specified herein, the examinations and tests listed in TABLE III shall be performed in accordance with the test methods specified in MIL-E-16400.

4.5.1 Performance tests. The switchboards shall be subjected to a performance test, including all operating controls, to determine compliance with the performance parameters specified in 3.2 and 3.3.

4.5.2 Interchangeability. The requirements for interchangeability specified in MIL-E-16400 shall be demonstrated on two switchboards chosen at random from each lot of 50 or less.

4.5.3 Reliability testing. Reliability testing shall be performed in accordance with MIL-STD-781, Test Plan IV C. Test temperature shall be maintained at 50° Celsius (C). Vibration stress (continuous) shall be as specified in the Combined environments for shipboard equipment paragraphs of MIL-STD-781. Conditions and parameters during test shall be as follows:

Input signal levels:	low level ±6 ±1 volt direct current (VDC) high level 60 mA maximum
Input impedance:	low level 6 kohms high level 100 ohms or less
Output signal levels:	low level ±6 ±1 VDC high level ±24 VDC 60 mA maximum
Output source impedance:	low level 6 kohms high level 100 ohms or less

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4.5.4 Maintainability demonstration.

4.5.4.1 Maintainability equipment demonstration. The contractor shall perform a maintainability demonstration at the organizational level. The demonstration shall be performed by qualified technicians and shall be used to verify that the equipment corrective maintenance requirements are met.

4.5.4.2 Demonstrated conditions. The equipment maintainability requirements shall be demonstrated in accordance with MIL-STD-471 by replacement of subassemblies (modules, individual PC boards) and chassis-mounted electronic, electrical, electromechanical, and mechanical components or parts at the organizational level. Sixty candidate corrective maintenance tasks shall be identified (selected in accordance with Appendix A of MIL-STD-471). The following information shall be made available for each candidate task:

a. Designation of specific faulty part

b. Failure mode

c. Means of introducing fault (substitution of faulty part or simulation thereof)

The procuring activity or its authorized representative shall use the candidate tasks as a guide to select a sample of 20 tasks for the demonstration.

4.5.4.3 Subassembly (module) maintainability. The module maintainability requirements shall be demonstrated by replacement of piece parts at the organizational, intermediate, or depot-level, as appropriate. Fault selection and simulation for each repairable module shall be in accordance with the procedure for the equipment maintainability demonstration. A total of 20 faults shall be selected by the procuring activity for the module maintainability demonstration.

4.5.4.4 Accept or reject criteria.

4.5.4.4.1 Corrective maintenance. The accept or reject criteria for the demonstration of the corrective maintenance times for the demonstration shall be as specified in TABLE IV.

TABLE IV. Acceptance criteria.

	M_{ct}	M_{maxct}
Acceptance level	8	0
Sample size	20	20

Acceptance shall occur when the number of observed corrective maintenance task times which exceed the required value of each specified index (M_{ct}, M_{maxct}) is less than or equal to that shown in TABLE IV corresponding to each index for the specified confidence level. The duration of each task shall be compared to the required value(s) and recorded as greater than or lesser than each index. The actual value shall also be recorded. An accept decision can only be made when an accept decision is made for both M_{ct} and M_{maxct}. Accept or reject criteria for preventive maintenance shall be the capability to perform each preventive maintenance task with no degradation of system performance. If degradation of system operations exists, the preventive maintenance test will have resulted in a reject decision.

4.5.4.5 Preventive maintenance. The demonstration shall qualitatively demonstrate that on-line preventive maintenance for the switchboards shall not interfere with on-line operation.

4.5.4.6 Demonstration environment. The demonstration at the organizational level shall be performed in an environment similar to that in which the switchboard will be installed.

4.5.4.7 Technicians. The procuring activity reserves the right to provide and select the technicians to perform the maintainability demonstration.

4.5.4.8 Technical documentation. Technical documentation shall be limited to the technical manual and related maintenance documentation delivered with the switchboard.

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4.5.4.9 Rejection. Failure (see 6.4) to conform to any of the maintainability requirements shall be cause for rejection of the demonstration. If a reject decision is reached, the procuring activity shall be immediately notified. The contractor shall, at no additional cost to the Government:

a. Develop an approach for redesign or correction of all deficiencies

b. Upon approval of a documented approach, the contractor shall repeat the demonstration until an accept decision is reached.

4.5.4.10 Acceptance. The maintainability demonstration shall exhibit that the switchboards conform to the accept criteria prior to any delivery.

4.5.5 EMI test. The requirements of 3.4.2 shall be verified by tests performed in accordance with MIL-STD-462. For Group C inspection, only tests CE03 and RS03 shall be performed.

4.5.6 Leakage current test.

WARNING

THIS TEST MAY BE HAZARDOUS DUE TO THE UNGROUNDED CONDITION OF THE EQUIPMENT DURING THE TEST. DO NOT TOUCH EXPOSED METAL SURFACES.

THE UNITED STATES GOVERNMENT NEITHER ASSUMES NOR ACCEPTS RESPONSIBILITY FOR ANY INJURY OR DAMAGE THAT MAY OCCUR DURING OR AS A RESULT OF THIS TEST.

4.5.6.1 Equipment connections. Each equipment directly connected to an external power source and units deriving power from the equipment shall be placed on an insulated surface. All safety ground conductors between the equipment and units deriving power from the equipment shall be intact. The safety ground conductor between the equipment under test and the source power shall be opened during the test. The equipment shall be connected as shown in FIGURE 2 if it is connected to single-phase power. When DC measurements are performed, as shown in FIGURE 3, do not reverse polarity unless the equipment has reverse polarity protection.

4.5.6.2 Measurement. Leakage current shall be measured on the equipment in its normal operating configuration and under maximum radio frequency loading conditions. Controls in each operating mode shall be such that maximum power will be utilized during leakage current measurements. The leakage current shall be determined by the voltage-drop method. A true root-mean-square voltmeter shall be used. The voltage measured across the 1500-ohm resistor shall not exceed 7.5 V at the highest nominal power line voltage and the highest and lowest nominal power line frequencies for which the equipment is designed. The overall measurement error shall not exceed five percent. The probe shall be used on all external conducting parts such as case, connector housings, recessed calibration or adjustment controls, and control shafts with knobs removed, and the voltage measured for every combination of switch positions available in the test diagram. The open safety ground conductor shall be reconnected immediately after the test is completed.

4.5.7 Workmanship.

4.5.7.1 General workmanship. The switchboard, including subassemblies and assemblies, shall be examined for workmanship and soldering during the fabrication and assembly process for conformance to the requirements of 3.1.

4.5.7.2 Workmanship screen. Prior to conducting the equipment conditioning test, vibration shall be performed on each switchboard. The vibration shall be performed at the module or drawer level. All hardware, including cables and connectors, shall be exposed to vibration.

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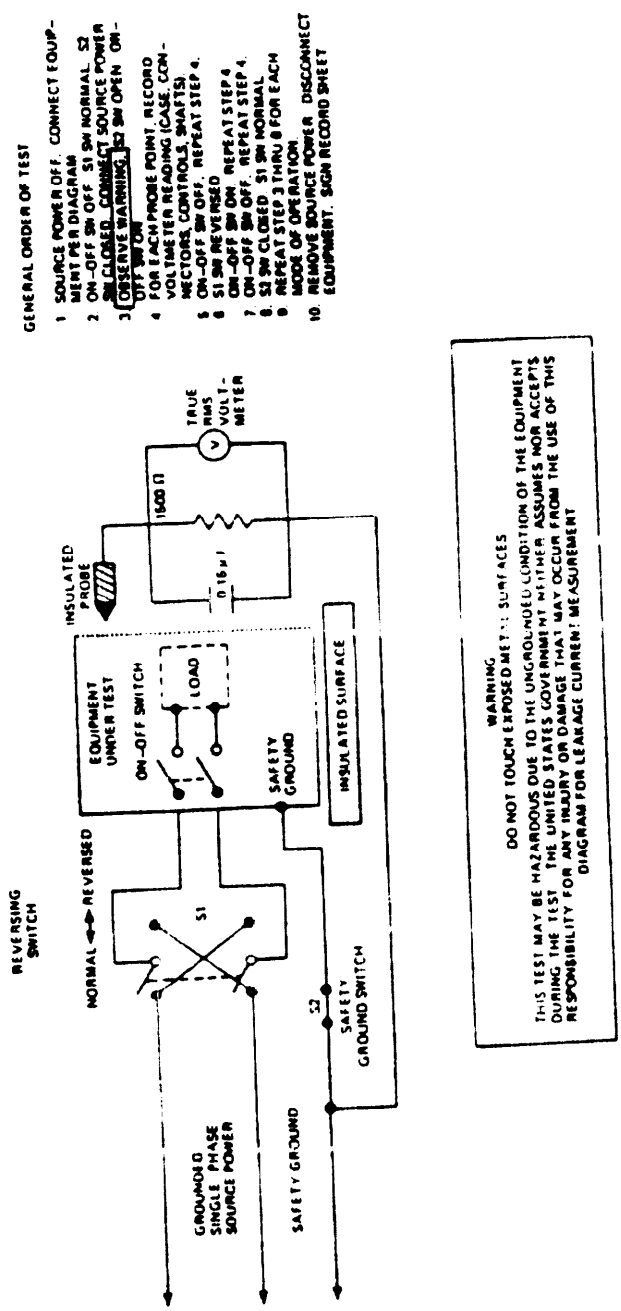


FIGURE 2. Single-phase or DC test diagram for leakage current measurement.

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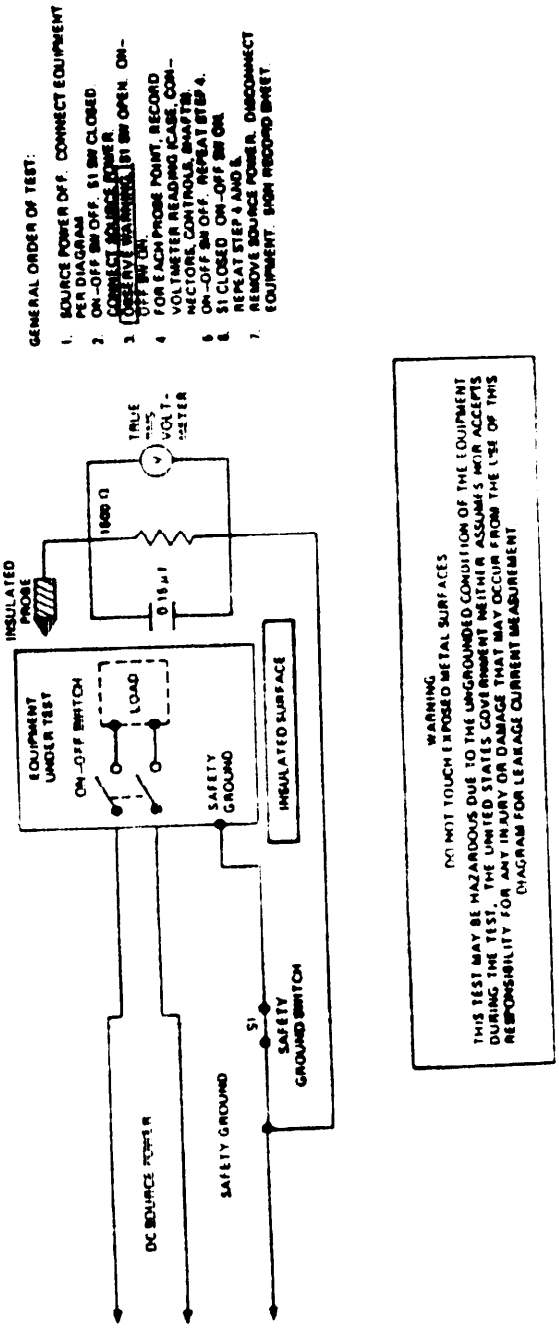


FIGURE 3. DC test diagram for leakage current measurement.

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4.5.7.2.1 Vibration. The vibration shall be random, or subject to procuring activity approval, pseudo-random or complex waveform vibration, for 10 minutes on the axis deemed most susceptible to vibration excitation. All test items shall be hard-mounted (without shock isolators) and shall be subjected to the vibration conditions of FIGURE 4. The control accelerometer shall be located next to one of the mounting points of the item under test. Equipment having a bandwidth no greater than 10 Hz for vibration frequencies up to 500 Hz and 100 Hz for vibration frequencies above 500 Hz shall be used for the control and analysis of the acceleration spectral density.

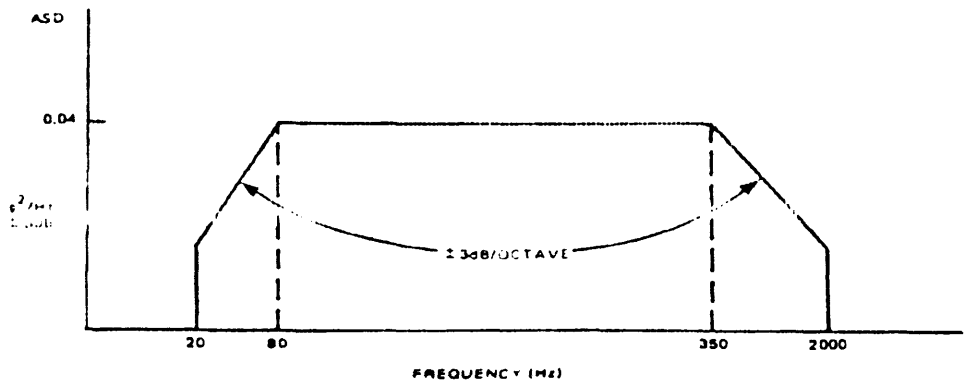


FIGURE 4. Random vibration curve.

The test item shall be energized during vibration and appropriate input signals applied to observe any abnormal conditions of the output functional characteristics. All failures occurring during test shall be corrected and the test repeated.

4.5.7.2.2 Temperature cycling. Each equipment shall be subjected to the temperature curve shown in FIGURE 5. The number of cycles required shall be 10 cycles. The temperature rate of change shall be not less than 5° C per minute. Equipment power shall be turned on and off at the indicated times. When practical, the equipment drawers, panels, and enclosures shall be opened or removed for maximum exposure to the changing temperature. The last temperature cycle shall be failure-free. When performance measurements are required, a functional operating test shall be performed. The dwell time shall be 80 percent of the time required for the largest electrical or electronic part to become temperature stabilized. Temperature stabilization shall be as defined in MIL-STD-810 for the non-operating condition.

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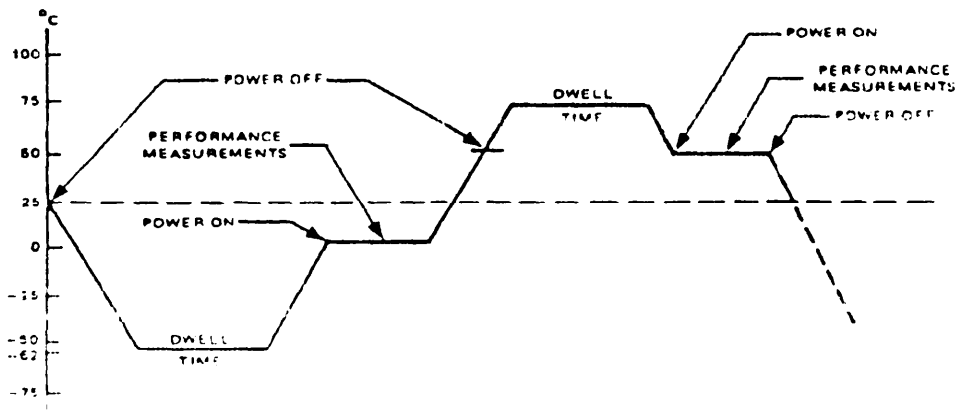


FIGURE 5. Temperature curve (one-cycle).

4.6 Inspection of preparation for delivery. Inspections shall be conducted to ensure conformance to the requirements of Section 5.

5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in Section 2 do not apply unless specifically stated in the contract. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order.)

5.1 Preservation, packaging, packing, and marking. Unless otherwise specified herein, preparation for delivery shall be in accordance with the applicable levels of preservation, packaging, packing, and marking specified in MIL-E-17555 (see 6.2.1).

6. NOTES

6.1 Intended use. The equipment covered by this specification is intended for use in submarines.

6.2 Ordering data.

6.2.1 Procurement requirements. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. Number of first article samples to be submitted if other than specified in 4.3

c. When reinspected production control and environmental sample units may not be accepted (see 4.4.1)

d. Levels of preservation, packaging, packing, and marking required (see 5.1)

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6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of Defense Acquisition Regulation (DAR) 7-1049 (n)(2) are invoked and the DD Form 1423 is not used, the data specified below will be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs:

Paragraph	Data Requirement	Applicable DID
4.4	Procedures, Test	UDI-T-22710
4.4	Test Report	DI-T-5247
4.5	Report, Thermal Survey	DI-R-7036
4.5.3	Reports, Reliability Test And Demonstration	DI-R-7034
4.5.3	Procedures, Reliability Test	DI-R-7035
4.5.4	Plan, Maintainability Demonstration	DI-R-2129
4.5.5	Report, Test, Electromagnetic Interference/Electromagnetic Compatibility (EMI/EMC)	DI-R-2066/R021

(DIDs related to this specification, and identified in Section 6, will be approved and listed as such in DoD 5000.19L., Vol. II, AMSL. Copies of DIDs required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 First article. When a first article is required it shall be tested and approved under the appropriate provisions of 7-104.55 of the DAR. The first article should be a first production item. The first article should consist of one unit. The contracting officer should include specific instructions in all procurement instruments, regarding arrangements for examinations, tests, and approval of the first article.

6.4 Failure. Failure shall be any malfunction or parameter deviation that prevents the switchboard from performing within the operational requirements specified herein.

6.5 Special tools. Special tools are defined as those tools not listed in the National Supply Catalog (copies of this catalog may be consulted in the office of the Defense Contract Administration Service).

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Preparing activity:
NAVY - EC
(Project 5815-N115)

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	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
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c. Reason/Rationale for Recommendation:

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7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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